

FX INTERVENTIONS IN A SMALL OPEN ECONOMY: THE CASE OF DOMESTIC NON-DELIVERABLE FORWARDS

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November 18, 2025

6th Biennial Conference on Financial Stability, Banco de México

INTRODUCTION

- ▶ Before the 2016 US Presidential Election, Bank of Mexico had already used significant reserves to reduce exchange rate volatility caused by the 2015 oil shock
- ▶ After Trump's victory, the peso vis-à-vis the US dollar experienced a large depreciation and high volatility
- ▶ The Foreign Exchange Commission (FEC) refrained from selling foreign exchange reserves due to potential risks
- ▶ They introduced Domestic Non-Deliverable Forwards (DNDF) as an alternative tool to mitigate the volatility of the peso
- ▶ **Research questions:**
 - ① How effective was this type of intervention employing derivatives in achieving the FEC goals?
 - ② Can this type of intervention prevent a self-fulfilling crisis or reduce the severity when it occurs?

▶ FX reserves and ER time series

IN THIS PAPER

- ✓ Using **high-frequency data**, we estimate the impact of this type of intervention on the **level** and **option-implied volatilities** of the Mexican peso
- ✓ We **separate** the effect by **announcements** and **auctions/implementations**
- ✓ We **explore** if there are any changes in **private sector's expectations** after the intervention
- ✓ We **rationalize** the empirical findings by **introducing insurance provision** into a standard Small Open Economy (SOE) model with collateral constraints à-la **Bianchi and Mendoza (2010)** and **financial intermediaries**

DNDF PROGRAM DESCRIPTION AND HOW IT WORKS

- ▶ From the [press release](#), the following are some [characteristics](#) of the intervention:
 - ✓ The program can size up to [20 billion US dollars](#)
 - ✓ The maximum tenor of the NDF's will be [12 months](#)
 - ✓ Only [local banks](#) that can operate derivatives can submit tenders for these instruments
 - ✓ Bank of Mexico will [roll over](#) the total amount outstanding of the NDFs until the FEC deems it [necessary](#)
- ▶ In terms of how [DNDF works](#): Let f_t be the forward rate determined by the central bank and domestic agents via an auction, e_{t+h} the USD/MXN exchange rate after h -periods, and q_t the amount of dollars/ "insurance"
- ▶ The [payment](#) that the domestic agent receives after h -periods is given by:

$$\text{Payment}_{t+h} = \begin{cases} (e_{t+h} - f_t) \times q_t > 0 & \text{if peso } \text{depreciates}, \\ (e_{t+h} - f_t) \times q_t < 0 & \text{if peso } \text{appreciates} \end{cases}$$

LITERATURE REVIEW AND CONTRIBUTION

- ① Effectiveness of interventions using non-deliverable forwards
Garcia and Volpon (2014), Chamon et al. (2017), Gonzalez et al. (2021), Macalós (2023)

Contribution: Provide evidence on the effect that policy had on the Mexican peso disentangling announcement vs. implementation

- ② Multiplicity of equilibria in SOE models and the role of policy tools
Bianchi (2011), Schmitt-Grohé and Uribe (2021), Davis et al. (2023), Bianchi and Coulibaly (2024)

Contribution: Show that this type of interventions can be used to avert a self-fulfilling financial crisis

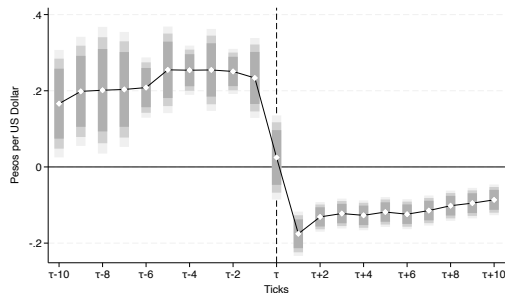
REDUCED-FORM EVIDENCE

EFFECT ON THE LEVEL OF THE PESO

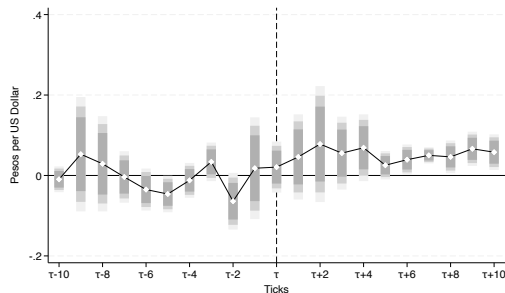
► Using [intraday](#) data on the USD/MXN, we estimate the following [linear model](#):

$$y_t = \alpha + \rho t + \sum_{s=-\tau}^{\tau} \beta_s \mathbb{1}\{t = T_0 + s\} + \theta VIX_t + \varepsilon_t,$$

FIGURE. Effect of DNDF on USD/MXN: First Announcement and Implementation



(A) Announcement



(B) Implementation

Notes: Program was announced at 2:14 pm on February 21, 2017. The first auction of this type of forwards occurred at 10:00 am on March 6, 2017. Gray bars represent CI at 80%, 90%, and 95%, respectively, constructed using HAC robust standard errors.

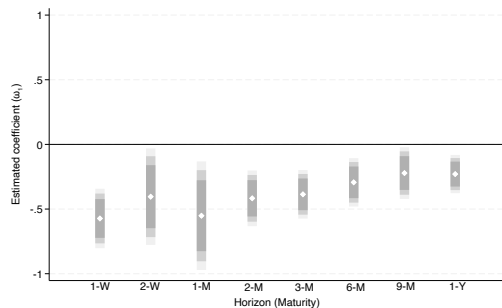
► [Data description](#) ► [Other announcements and auctions](#)

EFFECT ON OPTION-IMPLIED VOLATILITY OF THE PESO

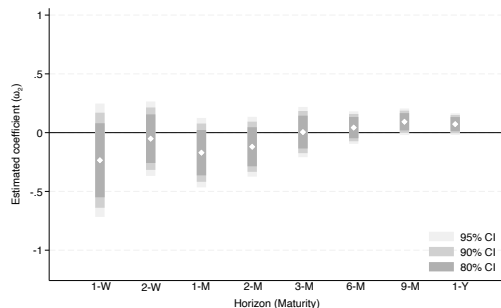
► Using **daily** data on option-implied volatilities, we estimate the following **linear model**:

$$y_{i,t} = \alpha_i + \gamma t + \rho y_{i,t-1} + \omega_1 \mathbb{A}\{t = T_0\} + \omega_2 \mathbb{I}\{t = T_1\} + \mathbf{x}_t \boldsymbol{\beta} + \eta z_{t-1} + \varepsilon_{i,t},$$

FIGURE. Effect of DNDF on Option-Implied Volatility of USD/MXN



(A) Announcement ($\hat{\omega}_1$)



(B) Implementation ($\hat{\omega}_2$)

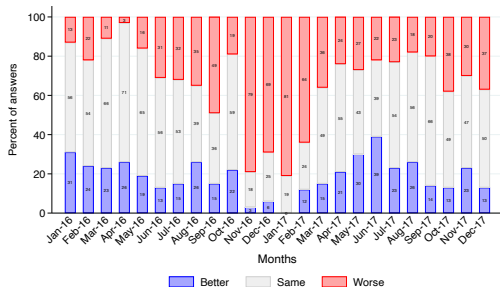
Notes: Indicator functions $\mathbb{A}$ and $\mathbb{I}$ take the value of 1 if in day t there was an announcement or auction, respectively, and 0 otherwise. For global controls $\mathbf{x}_t$, we include the VIX index, oil prices, and daily changes in Fed funds rate. For local controls z_{t-1} , we only use lagged changes in the monetary policy rate. Gray bars represent CI at 80%, 90%, and 95%, respectively, constructed using HAC robust standard errors.

► Panel estimates ► Synthetic control estimates

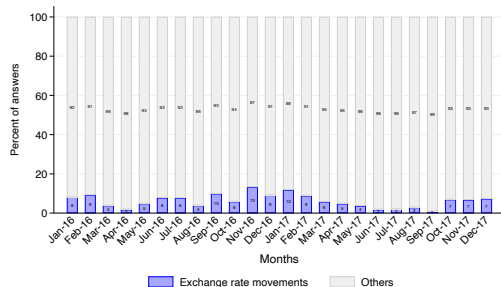
EXPECTATIONS OF THE PRIVATE SECTOR

- Survey data suggests a change in expectations after first announcement

FIGURE. Percent of Responses from Private Sector Specialists



(A) Business climate



(B) Limiting factors for growth

Notes: First panel display the percent of responses to the following question: “How do you think the business climate for productive activities of the private sector will evolve in the next six months compared to the last six months?”. The second panel displays the percent of responses to the following question: “In your opinion, during the next six months, what would be the three main limiting factors to economic growth?”

TAKING STOCK

► We find that:

- ① The announcement of the policy had an effect on the level of the exchange rate
 - ✓ First announcement had largest effect relative to extension announcements
 - ✓ There is no statistically significant effect associated with the auctions
- ② The announcement of the policy had an effect on option-implied volatilities
 - ✓ Short-maturities are the most affected ones; however, pattern is non-monotonic
 - ✓ As with the level, there is no evidence of an effect associated with the auctions
- ③ Survey responses from private sector suggests a change in expectations
 - ✓ After first announcement, larger fraction of “better economic conditions”
 - ✓ After first announcement, less concern of movements in the exchange rate

STRUCTURAL MODEL

ENVIRONMENT

► **Household:** unit measure of identical agents that maximizes life-time utility subject to borrowing constraint

$$\begin{aligned} \max_{\{c_t, M_{t+1}, d_{t+1}\}_{t \geq 0}} \quad & \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \left(u(c_t) + v \left(\frac{M_{t+1}}{p_t} \right) \right) \right] \quad \text{with } \beta \in (0, 1), \\ \text{s.t.} \quad & p_t^T c_t^T + p_t^N c_t^N + \tilde{R}_{t-1} \frac{e_{t-1} d_t}{R} + M_{t+1} = p_t^T y^T + w_t h_t + \frac{e_t d_{t+1}}{R} + M_t + \Pi_t - T_t, \\ & c_t = \left(\phi (c_t^T)^{\frac{\gamma-1}{\gamma}} + (1-\phi) (c_t^N)^{\frac{\gamma-1}{\gamma}} \right)^{\frac{\gamma}{\gamma-1}}, \text{ with } \gamma > 0 \\ & \frac{e_t d_{t+1}}{R} \leq \kappa (p_t^T y^T + w_t h_t), \text{ with } \kappa \in (0, 1) \end{aligned}$$

► **Sunspot shock:** Nonfundamental shock to agents beliefs on future exchange rates:

$$v_{t+1} = e_{t+1} - \mathbb{E}_t[e_{t+1}] \text{ with } v_{t+1} \sim \mathcal{F}$$

✓ Main source of collective pessimism and deleveraging that leads to currency depreciation

► **Nontradable production:** Linear technology in labor $y_t^N = h_t$ and there is DNWR (i.e., $w_t \geq w_{t-1}$)

ENVIRONMENT (CONT.)

► **Financial intermediaries:** homogeneous, **risk-averse** institutions that intermediates **external debt** from foreign lenders to households via a **zero net-capital strategy** and engages in a **forward contract** with Central bank (CB)

$$\begin{aligned} \max_{d_{t+1}, \theta_t \in [0,1]} & \mathbb{E}_t[\Pi_{t+1}] - \frac{\omega}{2} \text{Var}_t[\Pi_{t+1}], \\ \text{s.t. } & \Pi_{t+1} = \tilde{R}_t e_t \frac{d_{t+1}}{R} \left((1 - \theta_t) + \theta_t \frac{e_{t+1}}{f_t} \right) - e_{t+1} d_{t+1}, \end{aligned}$$

if $\frac{e_{t+1}}{f_t} \geq 1$ intermediaries get a **transfer in local currency** from CB; otherwise, they have to **pay to CB**

► **Central bank:** sets **money supply** M_{t+1} and determines **forward rate** f_t such that it satisfies an “equity” constraint:

$$\frac{a_{t+1}}{R} = \max \left\{ a_t - \left(\frac{1}{f_{t-1}} - \frac{1}{e_t} \right) \theta_{t-1} \tilde{R}_{t-1} e_{t-1} \frac{d_t}{R}, \underline{a} \right\},$$

CB **lump-sum transfers** T_t revenues from ΔM_{t+1} to households **net of resources** used to **replenish** “equity” if it falls below $\underline{a}$

EQUILIBRIUM ANALYSIS

- ▶ Under certain **calibrations**, this class of models feature **multiple equilibria** in which the economy is prone to **self-fulfilling crises** (see **Schmitt-Grohé and Uribe (2021)**)
 - ✓ These **bad equilibria** are characterized by **sharp depreciations**, current account **reversals**, and drop in **domestic absorption**
- ▶ Can the **intervention** using **forward contracts** reduce the likelihood of such **bad equilibria**?
 - ✓ **Yes!**
- ▶ To see this, consider the following expressions for the **borrowing limit**:

Aggregate borrowing limit in SS

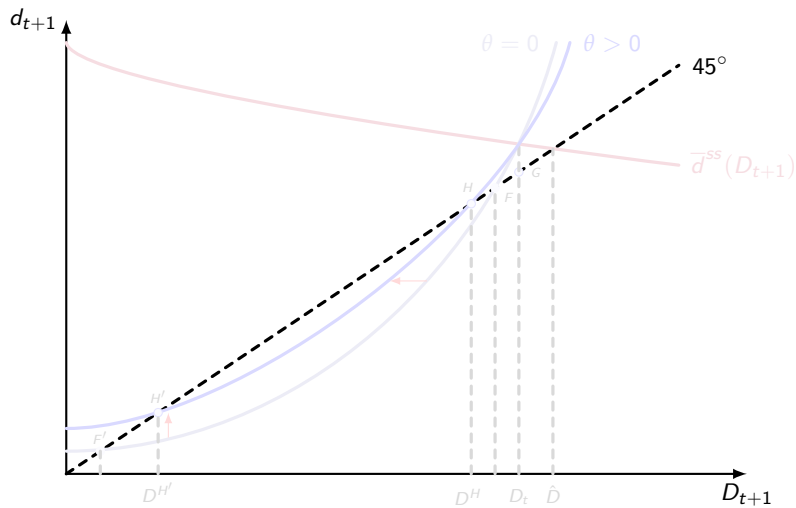
$$\bar{d}^{ss}(D_0) \equiv \kappa R \left(y^T + \left(\frac{1-\phi}{\phi} \right)^\gamma \left(\frac{w_{-1}}{e} \right)^{1-\gamma} \left(y^T - \frac{R-1}{R} D_0 \right) \right) \text{ with } \bar{d}^{ss'}(D_0) < 0$$

Individual borrowing limit with policy instrument

$$\bar{d}(D_{t+1}; D_t) \equiv \kappa R \left(y^T + \left(\frac{1-\phi}{\phi} \right)^\gamma \left(\frac{w_t}{e_t} \right)^{1-\gamma} \left(y^T + \frac{D_{t+1}}{R} - D_t + \frac{e_t - e}{e_t} \theta D_t \right) \right) \text{ with } \bar{d}' > 0 \text{ and } \bar{d}'' > 0$$

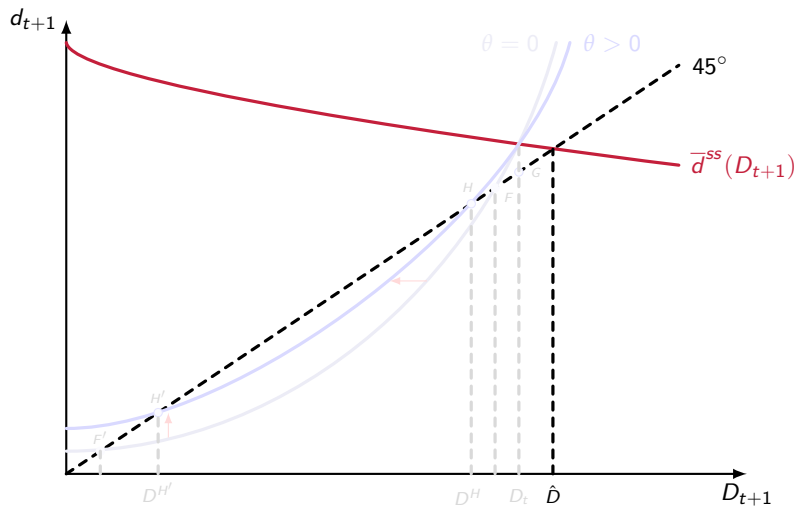
EQUILIBRIUM ANALYSIS (CONT.)

FIGURE. Multiplicity of Equilibria and Effect of Intervention



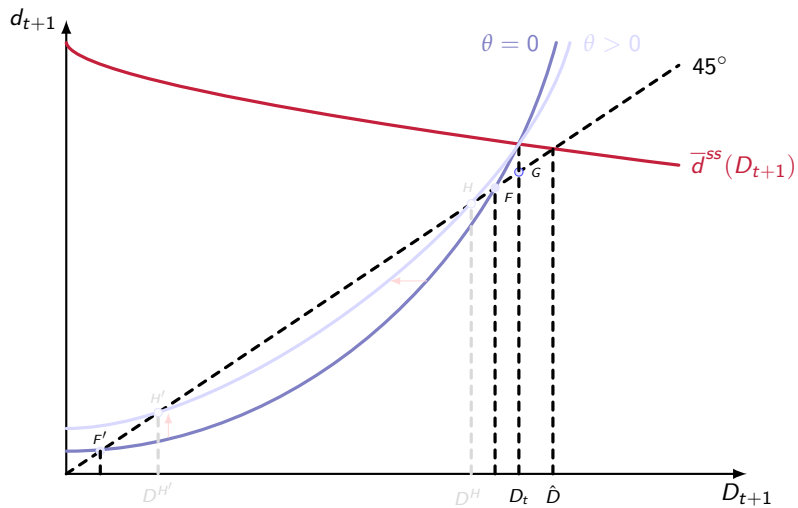
EQUILIBRIUM ANALYSIS (CONT.)

FIGURE. Multiplicity of Equilibria and Effect of Intervention



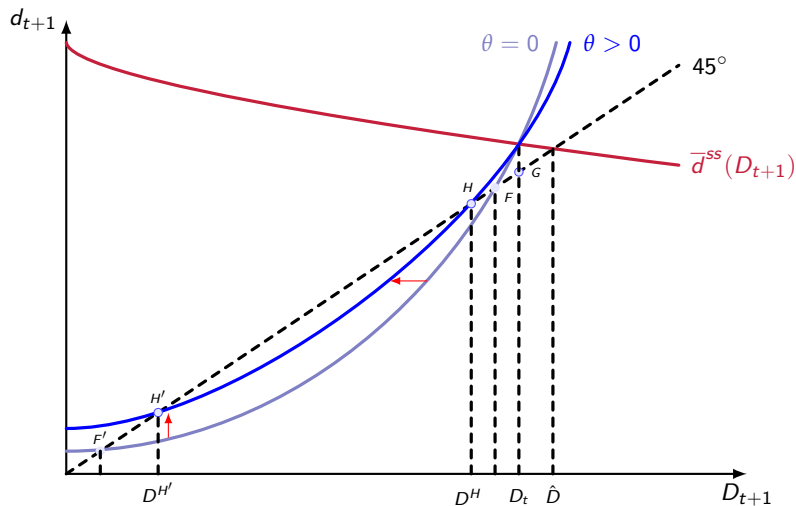
EQUILIBRIUM ANALYSIS (CONT.)

FIGURE. Multiplicity of Equilibria and Effect of Intervention



EQUILIBRIUM ANALYSIS (CONT.)

FIGURE. Multiplicity of Equilibria and Effect of Intervention



CONCLUSIONS

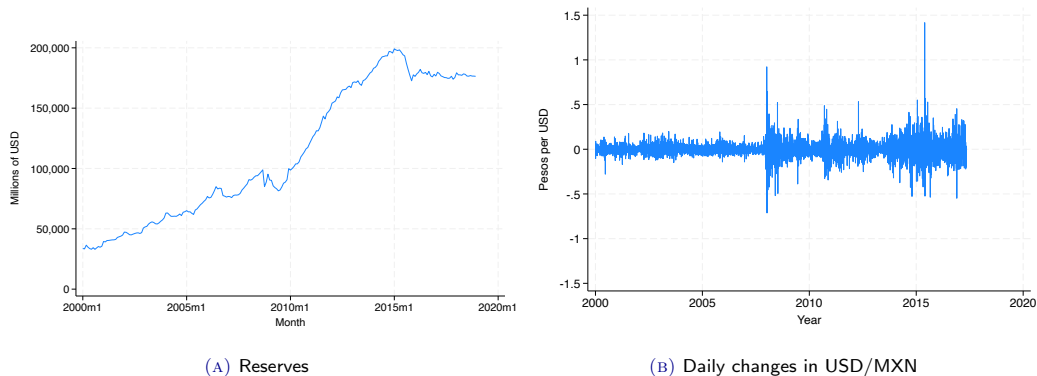
- ▶ We find **empirical evidence** on the effects of **DNDF policy** on MXN/USD exchange rate:
 - ✓ The peso **appreciated** in 32.90 cents per US dollar due to announcement
 - ✓ Option-implied volatilities **fall** 40 cents per US dollar on average due to announcement
 - ✓ Higher **effectiveness** of the policy **announcement** on **shorter** maturities
 - ✓ **No effect** associated with the **auctions**
 - ✓ **Survey responses** from private sector **suggests** a change in **expectations**
- ▶ We **rationalize** the empirical evidence using a **SOE model** with borrowing constraints that feature **multiple equilibria**
 - ✓ This forward contracts can **reduce the likelihood** of a crisis equilibrium
 - ✓ If crisis **materializes**, severity of contraction is **milder**
- ▶ **In progress:** Simulation exercises, analysis of **optimality** of this type of interventions, and **comparison** to other **tools** (e.g., foreign reserves and capital controls)

THANK YOU!

APPENDIX

FX RESERVES AND EXCHANGE RATE TIME SERIES

FIGURE. Evolution of reserves and USD/MXN exchange rate



Notes: The first panel, displays the monthly evolution of FX reserves reported by Bank of Mexico. The second panel, display the daily changes in the USD/MXN exchange rate.

BANK OF MEXICO DNDF ANNOUNCEMENT

► On February 21, 2017, the FEC announced the introduction of DNDF. Based on the official press release, the characteristics of the program are:

- ① The program can size up to **20 billion US dollars** taking into consideration the total nominal amount outstanding
- ② The maximum tenor of the NDF's will be **12 months**
- ③ Since the settlement of the forward is **non-deliverable**, the stock of international reserves is **unaffected** by the sale of these instruments
- ④ Only **local banks** that can operate derivatives can present tenders for these instruments
- ⑤ **Auctions** will be interactive, will last two minutes, and the allotment will be done at multiple price
- ⑥ Bank of Mexico will **roll over** the total amount outstanding of the NDFs until the Foreign Exchange Commission deems it **necessary**
- ⑦ The **first auction** will take place on March 6, 2017, for a total notional amount of one billion USD distributed along **six maturities**

► [Return](#)

DATA DESCRIPTION

► Intraday data:

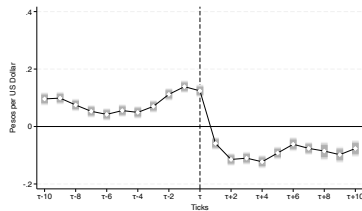
- ✓ Nominal exchange rate MXN/USD
- ✓ Controls: VIX index
- ✓ **Source:** Bloomberg
- ✓ **Period:** Days of announcements (Feb 21, Oct 25, and Dec 26, 2017) and implementations (Mar 6, Oct 26, and Dec 26, 2017)

► Daily and weekly data:

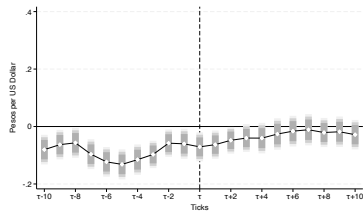
- ✓ Nominal exchange rate MXN/USD
- ✓ Implied volatilities of options on MXN/USD for different maturities (i.e., 1 week to 1 year)
- ✓ Controls: VIX index, oil prices, Fed Funds rate, and Bank of Mexico monetary policy rate
- ✓ **Source:** Bloomberg
- ✓ **Period:** Jan 1, 2016 to Dec 29, 2017

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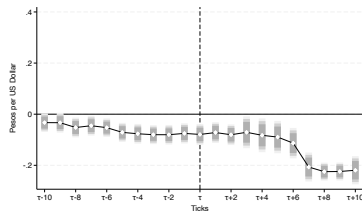
OTHER ANNOUNCEMENTS AND AUCTIONS



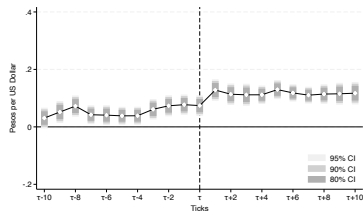
(A) Second Announcement



(B) Second Implementation



(C) Third Announcement



(D) Third Implementation

Notes: The diamond-shape marker corresponds to the point estimates of β_s , while the shaded areas correspond to the 80%, 90%, and 95% confidence intervals, respectively, constructed using heteroskedastic and autocorrelation robust standard errors.

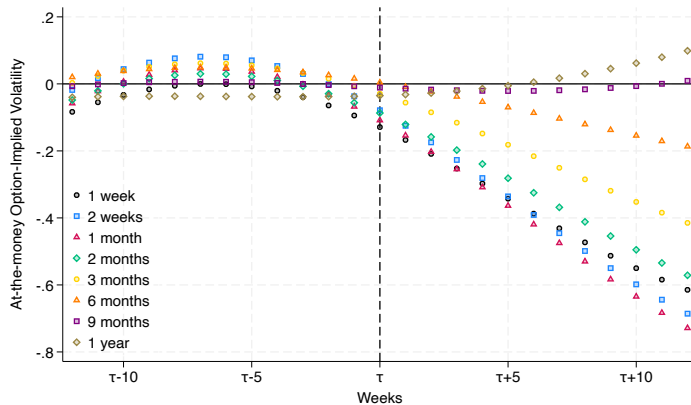
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PANEL ESTIMATES FOR OPTION-IMPLIED VOLATILITIES

Variable	Pooled			Panel		
	(1)	(2)	(3)	(4)	(5)	(6)
Lagged volatility	0.942*** (0.020)	0.929*** (0.021)		0.942*** (0.032)	0.929*** (0.032)	
Linear time trend	-0.0004*** (0.0001)	-0.0001 (0.0002)		-0.0004* (0.0002)	-0.0001 (0.0003)	
Announcement	-0.325*** (0.059)	-0.373*** (0.061)	-0.380*** (0.051)	-0.324*** (0.108)	-0.372*** (0.111)	-0.380*** (0.089)
Implementation	0.062 (0.051)	-0.043 (0.053)	0.018 (0.052)	0.062 (0.114)	-0.043 (0.111)	0.018 (0.114)
Lagged Change Mon. Pol.		-0.005*** (0.001)	-0.007*** (0.001)		-0.005** (0.002)	-0.007*** (0.002)
VIX Index		0.050*** (0.008)	0.028*** (0.007)		0.050*** (0.013)	0.028** (0.013)
Oil Prices (WTI)		0.019*** (0.004)	0.010*** (0.003)		0.019*** (0.006)	0.010* (0.006)
Daily Change in Fed Funds Rate		0.003** (0.002)	0.005*** (0.002)		0.003 (0.003)	0.005 (0.003)
Constant	✓	✓	✓	✓	✓	✓
Fixed Effects				✓	✓	✓
Obs.	4,160	4,160	4,160	4,160	4,160	4,160
R^2	0.903	0.906	0.013	0.902	0.905	0.013

Note: The R^2 associated with the panel data exercises corresponds to the within goodness-of-fit measure. ***, **, *, denote statistical significance at 1%, 5%, and 10%, respectively. When we use pooled data, we compute HAC robust standard errors and, when the model is estimated exploiting the panel structure, we compute [Driscoll and Kraay \(1998\)](#) standard errors to account for potential cross-sectional correlation.

SYNTHETIC CONTROL ESTIMATES

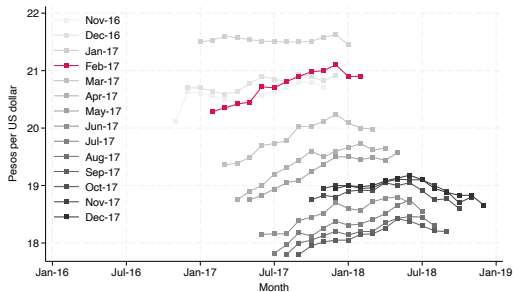


Notes: The markers correspond to the estimated treatment effect (i.e., the difference between the observed and synthetic implied volatility). t corresponds to the week in which the policy was announced by the Bank of Mexico. For almost all horizons h the countries with strictly positive weights are Brazil, Malaysia, and Turkey.

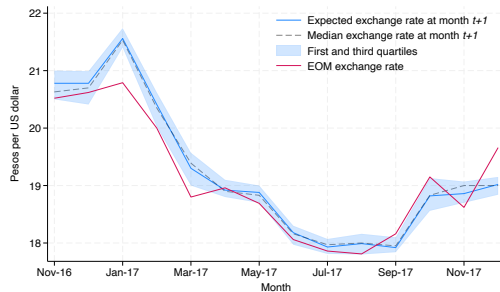
FIGURE. Effect over ATM Implied Volatilities

EXPECTATIONS OF THE EXCHANGE RATE

FIGURE. Evolution of expected exchange rate



(A) Monthly expectations t to $t+12$



(B) EOM vs. $\mathbb{E}_t[e_{t+1}]$

Notes: On the first panel, for each month t , we display the median of the forecast value of the exchange rate at period t and for the next 12 months from t reported by specialists in the private sector. On the second panel, the red line represents the monthly evolution of the peso-dollar exchange rate, the blue line corresponds to the expected value in month t of the exchange rate in period $t+1$, the black line is the median, and the blue area corresponds to the first and third quartiles.

COMPETITIVE EQUILIBRIUM

Equilibrium under DPDF Policy: Given initial conditions $\{f_{-1}, w_{-1}, d_0, a_0\}$, shocks $\{\nu_t\}_{t=0}^{\infty}$ and an arbitrary sequence of $\{f_t, M_t\}_{t=0}^{\infty}$, the **competitive equilibrium** is defined as a sequence of allocations $\{c_t^T, c_t^N, c_t, h_t, d_{t+1}, \Pi_t, \theta_{t+1}, \mu_t\}_{t=0}^{\infty}$, prices $\{e_t, p_t^N, p_t, w_t, \tilde{R}_t\}_{t=0}^{\infty}$, and $\{a_{t+1}, T_t\}_{t=0}^{\infty}$ such that:

- 1 Given initial debt d_0 , tradables endowment y^T , prices, and policy $\{f_t, T_t, a_{t+1}\}_{t=0}^{\infty}$, households maximize their life-time utility and firms maximize profit each period
- 2 Given the forward rate f_t and prices, financial intermediaries maximize their static profits Π_t
- 3 Given the initial asset a_0 and the forward rates $\{f_t\}_{t=0}^{\infty}$, and the external debt of the households $\{d_t\}_{t=0}^{\infty}$, the central bank asset a_{t+1} satisfies the “equity” constraint of the CB and T_t is set such that any revenues from ΔM_{t+1} net of resources used to replenish “equity” if it falls below $\underline{a}$ are transferred to households
- 4 Markets clear and resource constraint holds

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EQUILIBRIUM ANALYSIS: DETERMINISTIC STEADY-STATE

- ▶ We define a **deterministic** steady state when $\beta R = 1$ and **sticky wages**
- ▶ Agents start with **initial debt** D_0 , and central bank begins with **zero assets** ($\bar{a} < 0$)
- ▶ **Constant money supply** (zero inflation) and there is **no transfer** of resources from the central bank:

$$f = e \equiv \frac{\phi}{1 - \phi} \left(\frac{\bar{h}}{y^T - \frac{R-1}{R} D_0} \right)^{\frac{1}{\gamma}} w_{-1}$$

- ▶ **Borrowing constraint** does **not bind** at the steady state
- ▶ Steady-state debt limit is

$$\kappa R \left(y^T + \frac{p^N}{e} y^N \right) = \kappa R \left(y^T + \left(\frac{1 - \phi}{\phi} \right)^{\gamma} \left(\frac{w_{-1}}{e} \right)^{1 - \gamma} \left(y^T - \frac{R - 1}{R} D_0 \right) \right) \equiv \bar{d}^{ss}(D_0)$$

and is **decreasing** in D_0

▶ [Return](#)

EQUILIBRIUM ANALYSIS: MULTIPLE EQUILIBRIA UNDER DNDF

- Assume the CB picks **forward rates** f_t to be equal to future “steady-state” exchange rates such that

$$f_t(D_{t+1}) = \frac{\phi}{1-\phi} \left(\frac{\bar{h}}{y^T - (1-\beta)D_{t+1}} \right)^{\frac{1}{\gamma}} w_{t-1},$$

- Three results in this setting:

- ① Exchange rate **depreciates** ($e_t > e$) when $D_{t+1} < D_t$ so DNDF policy **boosts up** the debt limit

$$\bar{d}(D_{t+1}; D_t) = \kappa R \left(y^T + \left(\frac{1-\phi}{\phi} \right)^{\gamma} \left(\frac{w_t}{e_t} \right)^{1-\gamma} \left(y^T + \frac{D_{t+1}}{R} - D_t + \underbrace{\frac{e_t - e}{e_t} \theta D_t}_{\text{DNDF Settlement}} \right) \right)$$

- ② Agents should become **more pessimistic** for the crisis to **happen**
- ③ In the event of a crisis, the **most severe** contraction will be **less severe** than before the implementation of the policy

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